FYS5429/9429 March 5

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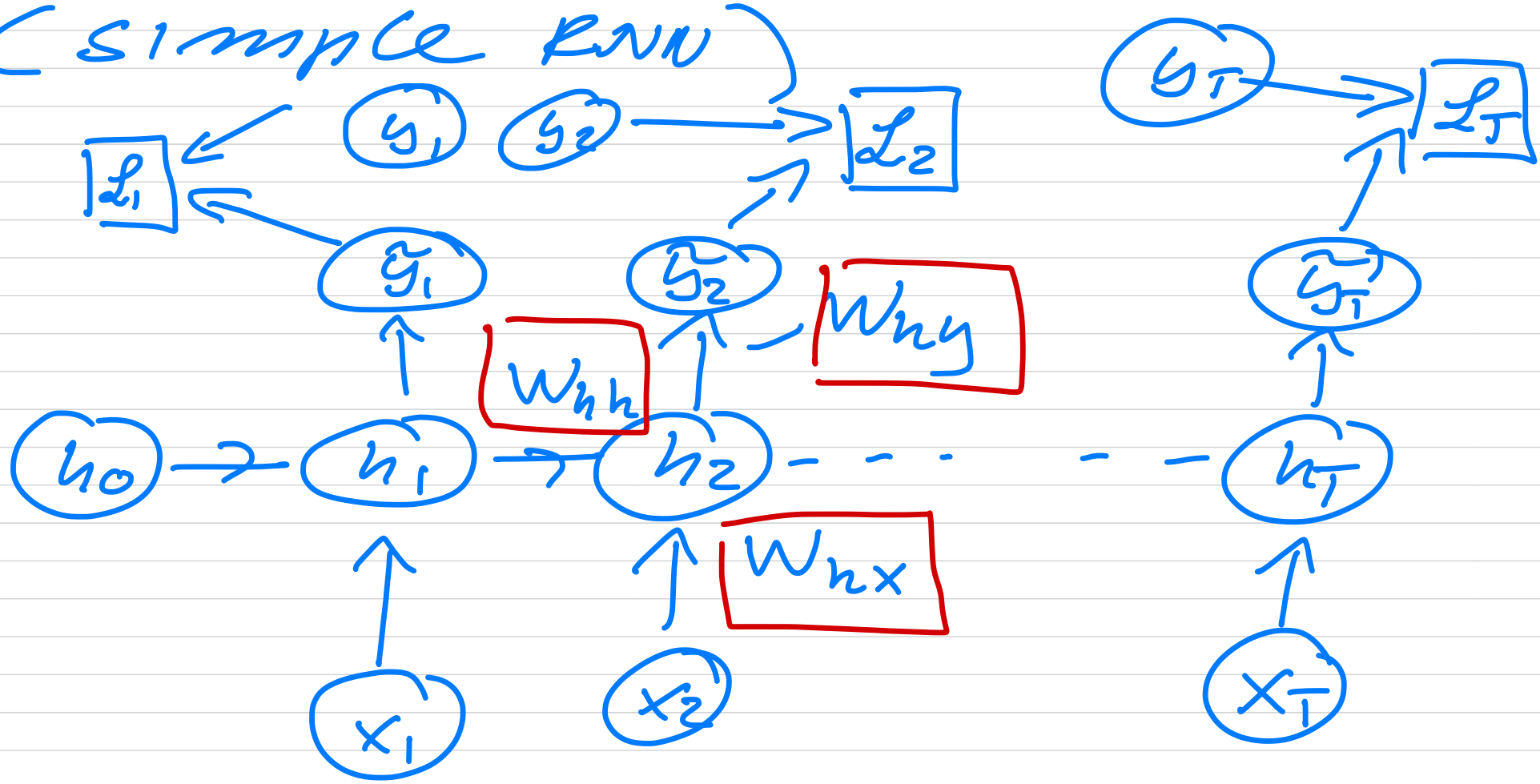
RNN and its basic math

(simple RNN)



many-to-many

input sequence

$$x_{1:T} \; (x_1, x_2, \ldots, x_T)$$

$$x_t \in \mathbb{R}^d$$

Typical predictions

- many-to-one
- one-to-many
- many-to-many

core challenge: capture dependencies across time steps t

An RNN maintains a hidden state $h_t \in \mathbb{R}^m$ that summarizes the past

$$h_t = f_\theta(x_t, h_{t-1})$$

parameters θ are shared across time steps

- short-range dependencies (local patterns)
- long-range dependencies

Training a basic (simple) RNN can be difficult

- vanishing gradients (long-range information dies out)

- exploding gradients (instabilities)

LSTMs/GRUs mitigate these issues

$$h_t = f(x_t, h_{t-1})$$

$$= \sigma(W_{hx} x_t + W_{hh} h_{t-1} + b_h)$$

output

$$y_t = \phi(W_{hy} \cdot h_t + b_y)$$

parameters

$$\theta = \{W_{hx}, W_{hh}, W_{hy}, b_h, b_y\}$$

$$a_t = W_{hx} x_t + W_{hh} h_{t-1} + b_h$$

$$h_t = \sigma(a_t)$$

$$x_t \in \mathbb{R}^d, \quad h_t \in \mathbb{R}^m$$

$$o_t = W_{hy} h_t + b_y$$

$$y_t = \phi(o_t)$$

$$W_{hx} \in \mathbb{R}^{m \times d} \qquad W_{hy} \in \mathbb{R}^{m \times k}$$

$$W_{hh} \in \mathbb{R}^{m \times m}$$

The recurrence couples time steps

$$h_t = \sigma(W_{hx} x_t + \boxed{W_{hh} h_{t-1}} + b_h)$$

- Forward pass computes $h_1, \ldots, h_T$

- Backward pass propagates sensitivities from T back to $\underline{1}$ (initialized h_0)

common objective

$$\mathcal{L} = \sum_{t=1}^{T} l_t(y_t, \tilde{y}_t)$$

Examples

- regression

$$l_t = \frac{1}{2} \| y_t - \tilde{y}_t \|_2^2$$

(norm-2 $\|x\|_2 = \sqrt{\sum_{i=1}^{n} x_i^2} = \sqrt{\langle x \rangle^2}$

- classification

$$l_t = - \sum_i \tilde{y}_{t;i} \log y_{t;i}$$

Training requires computing

$$\frac{\partial \mathcal{L}}{\partial \theta}$$

Since hidden states depend on earlier states, gradient must propagate through time. $\Rightarrow$ Backprop through time BPTT

RNN (simple) struggle with long-range dependencies

- exploding and vanishing gradient

Long-short term memory gates (LSTM) are able to cope with short-term and long-term

Correlations

Forget gate

$$f_t = \sigma(W_f x_t + U_f h_{t-1} + b_f)$$

input gate

$$i_t = \sigma(W_i x_t + U_i h_{t-1} + b_i)$$

candidate state

$$\tilde{c}_t = \tanh(W_c x_t + U_c h_{t-1})$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$$

output gate

$$o_t = \sigma(W_o x_t + U_o h_{t-1})$$